



ENGINEERED SOLUTIONS  
FOR A CLEANER, SAFER TOMORROW

RELIABLE SOLUTIONS.  
REAL PROGRESS.

# PISTON / PLUNGER DOSING SKID SYSTEM LNS-DZ-PDP

High-Pressure Chemical Injection & Additive Dosing System

DOSING  
INJECTION  
HIGH PRESSURE  
TRANSFER  
CONTROL

ROBUST  
RELIABLE  
CUSTOMIZABLE



PEOPLE  
INDUSTRY  
CLEANER  
TOMORROW



OIL & GAS



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PETROCHEMICAL



TERMINALS



INDUSTRIAL  
PROCESS

SOLUTIONS  
FOR A SAFER TOMORROW

[www.linesig.com](http://www.linesig.com)

## PRODUCT OVERVIEW & APPLICATION SCOPE

### 1. PRODUCT DESCRIPTION

The **LINESIG LNS-DZ-PDP** is a positive-displacement piston / plunger dosing skid engineered for precise chemical injection at medium to very high differential pressures. The pumping element provides fixed displacement per stroke, enabling accurate and repeatable delivery of additives, inhibitors, solvents, alcohols, treatment chemicals and selected hydrocarbons.

The system is particularly suitable where process backpressure is beyond the practical operating range of centrifugal pumps or where a compact, rugged high-pressure dosing package is required. Single or multiple pump trains can be configured with independent suction, pressure protection, pulsation control, metering and automation.

### 2. INDUSTRIES

- Oil & gas production, pipeline and terminal facilities
- Refineries and petrochemical plants
- Fuel blending and hydrocarbon additive systems
- Chemical processing and specialty chemical injection
- Power generation and boiler chemical treatment
- Utilities and industrial process plants
- High-pressure package systems and OEM skids

### 3. TYPICAL MEDIA

| Media Group           | Examples                                                                          | Engineering Consideration                                          |
|-----------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Hydrocarbon Additives | Corrosion inhibitors, drag-reducing additives, dyes, markers, lubricity improvers | Viscosity, lubricity, flammability and seal compatibility are key. |
| Alcohols / Solvents   | Methanol, ethanol and compatible solvents                                         | Hazardous-area and vapor-pressure considerations apply.            |
| Treatment Chemicals   | Antiscalants, demulsifiers, neutralizers, biocides                                | Materials selected by concentration and temperature.               |
| Specialty Chemicals   | Catalysts, inhibitors, injection chemicals                                        | Low-flow stability and high-pressure performance drive selection.  |

## TYPICAL PISTON / PLUNGER SKID CONFIGURATION

### 4. SYSTEM ARRANGEMENT

A standard **LNS-DZ-PDP** package may include a chemical day tank or remote suction source, suction strainer, isolation valves, piston/plunger metering pumps, pulsation dampeners, pressure relief valves, back-pressure devices, discharge check valves, pressure instrumentation, optional flow verification and a local or remote control panel.

| Subsystem               | Typical Arrangement                                                  |
|-------------------------|----------------------------------------------------------------------|
| Pump Configuration      | 1 x 100%, 1W+1S, 2W+1S or multi-line                                 |
| Suction                 | Common or individual manifold with strainer and isolation            |
| Pump Type               | Motor-driven reciprocating piston or plunger                         |
| Pulsation Control       | Discharge dampener; suction dampener if hydraulic conditions require |
| Overpressure Protection | Relief valve returning to suction/day tank or safe return system     |
| Flow Control            | Stroke length, speed control or combined method                      |
| Instrumentation         | Pressure, optional flow, tank level, local indicators                |
| Control                 | Manual, PLC/DCS, batch or ratio dosing                               |

### 5. PUMP TECHNOLOGY OPTIONS

- Piston metering pumps for clean, compatible liquid services
- Plunger metering pumps for rugged high-pressure injection
- Single-head or multi-head arrangements
- Simplex, duplex or multiplex drive assemblies
- Mechanically or hydraulically actuated configurations depending on duty

## TECHNICAL SPECIFICATIONS

### 6. GENERAL DESIGN DATA

| Parameter             | Typical / Available Basis                                                        |
|-----------------------|----------------------------------------------------------------------------------|
| Service               | High-pressure chemical dosing, process injection and additive feed               |
| Flow Range            | Project specific; typical package range from approximately 5 to 500 L/h per pump |
| Discharge Pressure    | Project specific; typically 10–200 bar, higher by engineered selection           |
| Pump Type             | Positive-displacement piston / plunger metering pump                             |
| Accuracy              | High repeatability when calibrated and operated within manufacturer range        |
| Turndown              | Pump/drive dependent; by stroke and/or speed control                             |
| Operating Temperature | According to fluid, seal and metallurgy selection                                |
| Wetted Materials      | SS316/316L, duplex, alloy materials; project specific                            |
| Packing / Seals       | PTFE, graphite, engineered polymer or service-specific packing                   |
| Skid Frame            | Coated carbon steel; stainless steel optional                                    |
| Connections           | ASME / EN flanged, threaded or tube connections as appropriate                   |
| Installation          | Indoor/outdoor; hazardous-area version optional                                  |

### 7. HYDRAULIC DESIGN BASIS

Pump selection shall be based on required minimum/normal/maximum flow, discharge pressure, suction pressure, fluid density, viscosity, vapor pressure, compressibility, gas entrainment, temperature and required turndown. NPSH and suction acceleration effects require particular attention in reciprocating pump systems.

## PULSATION, SUCTION & PRESSURE MANAGEMENT

### 8. PULSATION CONTROL

Reciprocating piston and plunger pumps inherently generate pulsating flow. Properly sized pulsation dampeners improve flow stability, reduce pressure ripple, protect instruments and piping, and lower vibration. Dampener sizing depends on pump displacement, number of heads, stroke frequency, allowable pressure variation and gas pre-charge.

### 9. SUCTION DESIGN

- Flooded suction is preferred where practical.
- Suction piping should be short, generously sized and free of unnecessary fittings.
- Acceleration head must be checked for high-speed or long suction runs.
- Provide strainers only where required and ensure low pressure drop.
- Suction pulsation dampeners may be used for demanding installations.
- Avoid vaporization or cavitation by maintaining adequate pressure margin above vapor pressure.

### 10. PRESSURE PROTECTION

| Device                       | Purpose                                                                   |
|------------------------------|---------------------------------------------------------------------------|
| Pressure Relief Valve        | Protects pump and discharge piping from blocked-line overpressure         |
| Back-Pressure Valve          | Stabilizes discharge and prevents uncontrolled siphoning where applicable |
| Check / Injection Valve      | Prevents process backflow                                                 |
| Pressure Gauge / Transmitter | Monitors pump discharge and provides alarm/trip input                     |
| Burst Disc (Optional)        | Secondary protection for selected critical duties                         |

## MATERIALS, PACKING & LEAKAGE CONTROL

### 11. MATERIAL SELECTION

| Duty                         | Typical Material Options                                     |
|------------------------------|--------------------------------------------------------------|
| Hydrocarbon / Fuel Additives | SS316L, duplex or compatible carbon steel/alloy              |
| Alcohol / Solvent Service    | SS316L with compatible packing and gaskets                   |
| Corrosive Chemicals          | Duplex, Hastelloy, Alloy 20 or selected specialty metallurgy |
| General Treatment Chemicals  | SS316L or project-specific alloy                             |

### 12. PLUNGER PACKING / SEAL SYSTEM

Packing selection is critical to performance and leakage control. Suitable materials may include PTFE-based packing, graphite composites or engineered polymers. For toxic, flammable or environmentally sensitive chemicals, leakage detection, closed drain, gland flushing or double containment may be incorporated.

### 13. LUBRICATION & DRIVE

Drive lubrication and gearbox requirements follow the selected pump manufacturer. Oil level, temperature and maintenance intervals shall be incorporated into the O&M philosophy. Variable-speed drives may be used where electronic capacity control is required.

## CONTROL, METERING & AUTOMATION

### 14. CAPACITY CONTROL

| Method                     | Typical Use                                |
|----------------------------|--------------------------------------------|
| Manual Stroke Adjustment   | Simple fixed duty                          |
| Electric Stroke Positioner | Remote capacity adjustment                 |
| VFD Speed Control          | Wide controllable flow range               |
| Combined Stroke + Speed    | Extended turndown / optimized accuracy     |
| Flow Feedback Control      | Closed-loop control using flowmeter        |
| Ratio Control              | Chemical flow proportional to process flow |

### 15. INSTRUMENTATION

- Discharge pressure gauge and/or transmitter
- Optional flowmeter or flow verification device
- Tank level transmitter / low-level switch
- Pump speed or stroke position feedback
- Relief-valve discharge monitoring where required
- Leak detection at packing/gland drain
- Local temperature or vibration monitoring for critical service

### 16. AUTOMATION

PLC/HMI or DCS integration may provide automatic duty/standby changeover, flow setpoint control, pressure alarms, low-level inhibit, motor trip, high-pressure shutdown, leak alarm, batch totalization, local/remote selection and ESD interface.

## SAFETY & HAZARDOUS AREA EXECUTION

### 17. SAFETY FEATURES

| Protection                   | Typical Function                                  |
|------------------------------|---------------------------------------------------|
| High Discharge Pressure Trip | Protects pump and downstream equipment            |
| Mechanical Relief Valve      | Independent overpressure protection               |
| Low Suction Level / Pressure | Prevents loss of prime or vaporization            |
| Packing Leak Detection       | Early detection of product leakage                |
| Emergency Stop               | Local and optional remote ESD                     |
| Bund / Drip Tray             | Contains leakage and facilitates drain collection |
| Earthing / Bonding           | Required for flammable service where applicable   |

### 18. HAZARDOUS SERVICE

For hydrocarbon, alcohol or solvent duty, motors, junction boxes, solenoids, transmitters and local control devices shall be selected according to the project hazardous-area classification. ATEX, IECEx or other certification requirements are defined at project stage.

### 19. PRESSURE CLASS

Piping, valves, instruments and fittings shall be selected from the maximum allowable working pressure and design temperature, not only the normal pump discharge pressure. Pressure class and relief settings shall be coordinated during detailed engineering.

## APPLICATION CONFIGURATIONS & FAMILY SELECTION

### 20. TYPICAL CONFIGURATIONS

| Configuration               | Recommended Use                                 |
|-----------------------------|-------------------------------------------------|
| 1 x 100%                    | Single high-pressure injection point            |
| 1W + 1S                     | Critical continuous injection with standby      |
| 2W + 1S                     | Two active injection points with common standby |
| Multi-Head Pump             | Multiple proportional streams from common drive |
| Multiple Independent Trains | Multi-point injection with independent control  |

### 21. WHEN TO SELECT LNS-DZ-PDP

- Injection pressure is high or very high.
- Accurate, repeatable positive-displacement dosing is required.
- The liquid is clean enough for reciprocating pump check valves and packing.
- A rugged mechanical solution is preferred for hydrocarbon or process additive injection.
- Low-to-medium flow at high pressure is more important than pulse-free delivery.
- The process requires ratio, batch or remote automated chemical injection.

### 22. LINESIG DOSING FAMILY

| Requirement                         | Recommended Model                 |
|-------------------------------------|-----------------------------------|
| Medium/high-flow smooth blending    | LNS-DZ-CEN - Centrifugal          |
| Corrosive / leak-sensitive metering | LNS-DZ-PDD - Diaphragm            |
| High-pressure process injection     | LNS-DZ-PDP - Piston / Plunger     |
| Very low-flow precision dosing      | LNS-DZ-PDG - Precision Micro Gear |

## ORDERING DATA & STANDARD SCOPE

### 23. MINIMUM DATA REQUIRED FOR FINAL SIZING

- Chemical/product name, concentration and safety classification
- Minimum / normal / maximum flow per pump
- Normal and maximum discharge pressure
- Suction pressure / liquid level range and source tank arrangement
- Density, viscosity and vapor pressure
- Minimum / normal / maximum temperature
- Required turndown and accuracy
- Pump redundancy philosophy and number of injection points
- Required material of construction and packing/seal preference
- Hazardous-area classification
- Control mode: local, ratio, batch, PLC/DCS or remote
- Applicable piping, electrical and inspection standards

## 24. TYPICAL STANDARD SCOPE

| Item | Typical Supply                                                                                  |
|------|-------------------------------------------------------------------------------------------------|
| A    | Skid structural frame and supports                                                              |
| B    | Piston / plunger metering pump(s) and drive                                                     |
| C    | Suction manifold, isolation and optional strainer                                               |
| D    | Pulsation dampener(s) as required                                                               |
| E    | Relief and back-pressure valves                                                                 |
| F    | Discharge check / injection valves                                                              |
| G    | Pressure instrumentation                                                                        |
| H    | Optional flowmeter / flow verification                                                          |
| I    | Leak collection / drain connections                                                             |
| J    | Local PLC/HMI or junction box when specified                                                    |
| K    | Internal piping, tubing and cable supports                                                      |
| L    | Documentation: datasheet, GA, P&ID, I/O list, control philosophy and test records as applicable |



**HIGH PRESSURE. PRECISE INJECTION. RELIABLE PROCESS CONTROL.**

***LINESIG LNS-DZ-PDP — Engineered for demanding chemical and hydrocarbon injection service.***